

Name: _____

Instructions:

- Log-in to your Google Classroom
- Add Google Slides and then place your answers inside it
- Read the questions in this paper carefully before answering it online

Learning Unit 1 [40 Marks]

Case Study: FreshMart Retail Pte. Ltd.

FreshMart Retail Pte. Ltd. is a traditional supermarket company operating five physical stores in Singapore.

The company recently launched an online shopping website. Customers can:

- Browse products online
- Add products to a shopping cart
- Make online payments
- Choose home delivery

However, FreshMart is facing several problems:

- Store inventory and online inventory are not fully integrated.
- Employees manually update some inventory information.
- Suppliers still receive many orders through email and telephone.
- Customers cannot see real-time delivery status.
- Customer purchasing data is not being fully analyzed.
- Competitors are offering personalized recommendations and mobile services.
- Customers increasingly expect faster delivery and seamless online/offline shopping.
- Management is considering using cloud computing, AI, Big Data Analytics and IoT.
- Some employees are concerned that automation and AI may change their jobs.

Management wants to transform FreshMart from a business that **simply sells products online** into a **holistic digital business and digital value network**.

Task 1: E-Commerce vs. Digital Business

Analyze FreshMart's current digital operations.

A. Identify 4 Traditional E-Commerce Activities

Identify **four activities** that represent traditional e-commerce.

For each activity, explain why it can be classified as traditional e-commerce.

Activity	Explanation
1.	
2.	
3.	
4.	

Traditional e-commerce in the unit is primarily focused on online buying and selling, digital storefronts, customer portals and payment integration.

B. Identify 4 Digital Business Opportunities

Recommend **four capabilities** FreshMart should introduce to become a more holistic digital business.

Examples could include:

- AI-powered recommendations
- Integrated inventory systems
- Cloud-based operations
- Automated supplier integration
- Data analytics
- IoT-enabled inventory management
- Digital customer ecosystem

Explain how each capability goes beyond simply selling products online.

The unit describes holistic digital business as integrating internal operations, supply chains, partners and customer touchpoints.

PART B – Digital Value Network

Task 2: From Linear Supply Chain to Digital Value Network

FreshMart currently operates largely through a traditional linear process:

Supplier → FreshMart → Customer

Redesign this model into a **digital value network**.

The unit contrasts a sequential supply chain with a multi-directional platform ecosystem, where customers can become active participants and data can flow continuously across the network.

Step 1 – Draw the Existing Linear Model

Create a diagram showing:

Suppliers → FreshMart Distribution/Stores → Customers

Add any other relevant traditional participants.

Step 2 – Create the Digital Value Network

Your new diagram must include at least:

- Customers
- Suppliers
- FreshMart
- Delivery partners
- Payment providers
- Digital platform
- Data/analytics
- Technology providers
- At least **two feedback loops**
- At least **two third-party partners**

Step 3 – Explain the Transformation

Answer:

1. How is the new model different from the original linear model?
2. What is the role of the customer?
3. How does data move through the network?
4. Where are feedback loops created?
5. How can the digital network create additional value?
6. How can software and network effects support scalability?

The source identifies differences in **structure, customer role, data flow and scalability** between linear models and digital value networks.

PART C – Digital Transformation Drivers

Task 3: Identify and Prioritize Transformation Drivers

FreshMart is experiencing changes in its market, technology and customer environment.

Identify:

- **3 Market Drivers**
- **3 Technology Drivers**
- **3 Customer Drivers**

The unit specifically categorizes transformation forces into these three groups. Market drivers include digital entrants and industry disruption; technology drivers include cloud, AI, Big Data Analytics and IoT; customer drivers include personalization, omnichannel access, instant fulfillment and transparent interactions.

PART D – Strategic Alignment & Digital KPIs

Task 4: Align Digital Strategy with Business Goals

FreshMart's management has identified the following business objectives:

1. Increase revenue.
2. Reduce operating costs.
3. Improve customer satisfaction.
4. Increase customer retention.
5. Improve operational efficiency.

Develop appropriate digital initiatives and KPIs.

The unit emphasizes connecting digital KPIs to enterprise goals such as **revenue, cost reduction and market share**.

Complete the Table

Business Goal	Digital Initiative	KPI	Target
Increase revenue			
Reduce costs			
Customer satisfaction			
Customer retention			

Learning Unit 2 [30 Marks]

You are appointed as a **Digital Business Strategy Consultant** for a traditional business that wants to transform its business model using digital technologies.

Your client has been operating successfully through traditional methods but is facing increasing competition from digitally enabled businesses.

The management team wants you to:

- Analyse the current digital environment.
- Identify competitive forces.
- Identify strengths, weaknesses, opportunities and threats.
- Identify opportunities to create new market space.
- Reconfigure the company's value chain and Business Model Canvas.
- Assess the company's digital maturity.
- Develop a practical digital transformation roadmap.

Choose ONE Business

Students may choose **one** of the following businesses:

Option A – Traditional Restaurant

Example:

A local restaurant currently relies mainly on walk-in customers, telephone reservations and cash/card payments.

Option B – Clothing Retailer

Example:

A physical clothing shop with a Facebook page but no e-commerce website.

Option C – Travel Agency

Example:

A traditional travel agency that mainly sells flights and holiday packages through its physical office.

Option D – Small Hotel

Example:

A 40-room independent hotel that receives bookings mainly through telephone, email and third-party booking platforms.

Option E – Grocery Store

Example:

A neighbourhood grocery store with a physical shop but no online ordering or delivery service.

Option F – Student's Own Business

Students may propose another business **with lecturer approval**.

PART A – Digital Environmental & Competitive Analysis

Students must conduct **three digital strategic analyses**.

Task 1. Digital SWOT Analysis

Create a Digital SWOT analysis for the selected organisation.

Students must provide **at least four items under each category**.

Strengths	Weaknesses
Internal digital capabilities	Internal digital limitations
Digital channels	Technology limitations
Data capabilities	Skills gaps
Customer experience	Process limitations
Opportunities	Threats
AI	Digital competitors
E-commerce	Cybersecurity
Digital platforms	Changing customer behaviour
Automation	New technologies

Task 2. Digital PESTLE Analysis

Analyse the external digital environment using:

- Political
- Economic
- Social
- Technological
- Legal
- Environmental

Students must identify **at least one relevant factor under each category**.

For each factor, explain:

1. What is changing?
2. Why is it relevant?
3. How could it affect the business?
4. Is it an opportunity or potential risk?

PART B – Reconfiguring the Digital Value Chain & BMC

Students must demonstrate how digital technology can change the organisation's existing business model.

Task 1. Current Value Chain

Map the organisation's **current value chain**.

Example:

Supplier

↓

Inventory

↓

Physical Store

↓

Sales Staff

↓

Customer

↓

After-sales service

Students should identify where digital technology is currently used and where manual processes exist.

PART C – Digital Blue Ocean & Value Innovation

Students must use the **Four Actions Framework** to create a new digital value proposition.

Task 1. Eliminate

Identify at least **three activities/factors that could be eliminated**.

Explain **why they can be eliminated**.

Task 2. Reduce – 4 marks

Identify at least **three factors that can be reduced**.

Examples:

- Waiting time
- Transaction costs
- Customer effort
- Administrative work
- Processing time

Task 3. Raise

Identify at least **three factors that should be increased**.

Examples:

- Convenience
- Personalisation
- Speed
- Transparency
- Customer engagement

Task 4. Create

Identify at least **three completely new sources of customer value** enabled by digital technology.

Examples:

- AI recommendations
- Digital loyalty ecosystem
- Personalised subscription
- On-demand service
- Online marketplace
- Predictive service
- Virtual consultation

Students must explain **how the new value could create a different market opportunity**.

Learning Unit 3 [30 Marks]

You are a **Digital Business Strategy Consultant**.

A traditional business wants to transform its operations and introduce a digital platform.

Choose **ONE** of the following businesses:

1. **Hotel / Tourism Company**
2. **Food Delivery Company**
3. **Retail / E-Commerce Company**
4. **Logistics Company**
5. **Education / Training Provider**
6. **Banking / FinTech Company**

The business currently relies heavily on traditional processes and wants to develop a **digital platform supported by APIs and digital transformation strategies**.

Your task is to propose how the business could become a digitally enabled business.

PART A – Multi-Sided Platform Architecture

Task 1

Design a digital platform for your selected business.

Step 1: Identify the platform participants

Identify at least **two sides** of your platform.

For example, for a food delivery platform:

Customers ↔ Platform ↔ Restaurants ↔ Delivery Partners

For a hotel platform:

Guests ↔ Platform ↔ Hotels/Hosts

Step 2: Explain the role of each participant

Create a table similar to:

Participant	Role	What do they want?	Value received
Customers	Purchase food	Convenient meals	Easy ordering
Restaurants	Sell food	More customers	Additional sales
Delivery partners	Deliver food	Income	Delivery opportunities
Platform	Connect participants	Revenue	Transaction fees

Step 3: Explain the network effects

Explain:

- What happens when more users join?
- How does one side benefit from the growth of another side?
- Are there any negative effects from having too many participants?

Step 4: Explain platform orchestration

Explain how your platform will manage:

- Matching
- Search
- Transactions
- Payments
- Ratings/reviews
- Customer communication
- Trust and safety

PART B – API Network & Ecosystem

Your platform should not build every technology from scratch.

Identify at least **FIVE APIs or external digital services** that your platform could integrate.

Example: Food Delivery Platform

API / Service	Purpose	How it is used
Payment API	Process payments	Customer pays for order
Maps API	Location/navigation	Driver finds customer
SMS API	Notifications	Customer receives order updates
Analytics API	Analyse behaviour	Platform studies customer activity
Authentication API	User verification	Secure login

Step 1: Explain Each API

For each API, explain:

1. What does it do?
2. Who uses it?
3. What information is exchanged?
4. What business benefit does it provide?

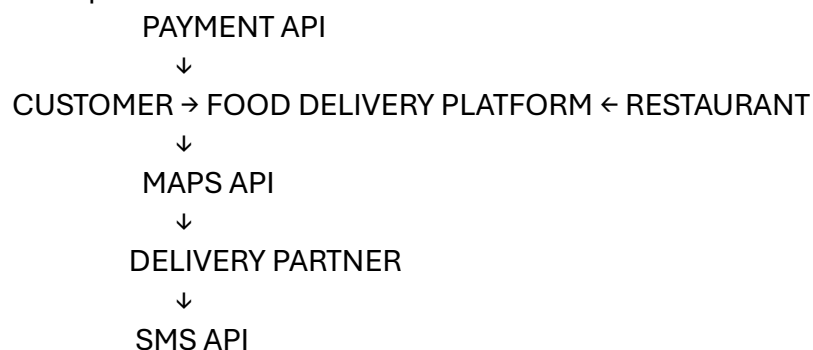
Example

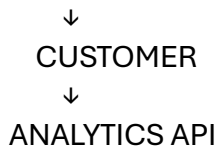
Maps API

The Maps API provides location and route information. The food delivery application uses it to identify the restaurant and customer location and help delivery partners find the fastest route.

Step 2: Create an API Ecosystem Diagram

Example:





Step 3: Explain Ecosystem Governance

Explain how your platform will manage:

- Authentication
- Data security
- User access
- Data privacy
- API usage
- Partner requirements
- Revenue sharing

PART C – Business Transformation Strategy

Your selected company is currently a **traditional/legacy enterprise**.
You must propose how it can transform into a digitally enabled organisation.

Choose at least **FOUR** strategies from the following:

1. Cloud Migration

Explain how cloud technology can improve the organisation.

2. AI-Driven Workflow Automation

Identify repetitive activities that AI/automation could perform.

3. Customer Experience Modernisation

Explain how digital technology can improve the customer journey.

4. IoT & Supply Chain Digitisation

Explain how connected devices/sensors can improve operations.

5. Data-Driven Business Model Innovation

Explain how data can create new products, services or revenue.

6. Agile ERP

Explain how different departments can be integrated through an ERP system.